Regional Differentials in Demographic and Health Indicators: Evidence from NFHS Data on Punjab and Jammu & Kashmir

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Group No: 1
Course: Bachelor of Statistics (Hons.)
Academic Year: 2025-2026

1 Introduction

India exhibits substantial regional heterogeneity in demographic and health outcomes, shaped by social norms, economic development, gender relations, institutional access, and historical trajectories. Even among states with relatively similar human development profiles, patterns of fertility behaviour, marriage timing, child health, and women's economic participation may differ sharply. In this study, we focus on two northern Indian states – Punjab and Jammu & Kashmir – and investigate state-level differentials across selected demographic indicators using nationally representative data from NFHS-5. The analysis examines five interconnected themes that reflect both population processes and gendered life-course transitions: fertility determinants, early marriage patterns, age at first birth, Child Survival Under Five, and women's employment. Each theme addresses a specific but complementary component of demographic behaviour, while also emphasising the role of socioeconomic stratification through education, wealth inequality, and rural–urban residence. Using survey-weighted statistical methods, regression-based modelling, and appropriate multiple-testing corrections, this report aims to generate comparative empirical evidence on how demographic outcomes in these two states vary, and to quantify the pathways through which socioeconomic factors shape these differences. Through this comparative lens, we seek to contribute to a more nuanced understanding of population dynamics and women's status in Punjab and Jammu & Kashmir.

2 Research questions

2.1 Determinants of fertility

How do education level, wealth status, and place of residence (urban or rural) influence the number of children per woman in Jammu & Kashmir and Punjab?

2.2 Early Marriage in India: Prevalence and Predictors

- What is the prevalence of early marriage (age at first marriage < 18) in Punjab and Jammu & Kashmir by education category?
- Is the prevalence higher in Punjab than in Jammu & Kashmir within the education strata? (Multiple-testing corrected.)
- What is the adjusted association of education, wealth and residence with early marriage (survey-weighted logistic regression)?

2.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

To what extent do urban–rural differences influence the age at first birth among women, and how do education level and household wealth contribute to explaining this difference in Jammu and Kashmir and Punjab?

2.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

What are the socioeconomic and demographic determinants of child survival among women aged 15–49 in the Indian states of Jammu & Kashmir and Punjab, and how do factors such as household wealth, maternal education, place of residence, and access to water and sanitation facilities influence the survival status of children?

2.5 Influence in employment of women

What socioeconomic factors (education, wealth, residence, and state) affect the employment of women in the Indian states of Punjab and Jammu & Kashmir ?

3 Research objectives

3.1 Determinants of fertility

The main objectives of this study are for the states of Jammu & Kashmir and Punjab:

- To investigate whether education significantly affects fertility behavior.
- To evaluate the effect of household wealth on the number of children.
- To examine whether fertility differs significantly between urban and rural populations.

3.2 Early Marriage in India: Prevalence and Predictors

- To quantify early marriage prevalence across education strata in Punjab and Jammu & Kashmir.
- To evaluate whether Punjab exhibits significantly higher early marriage prevalence relative to J&K after controlling for education.
- To estimate adjusted associations of early marriage with education, wealth, residence, and state using survey-weighted regression.

3.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

- To examine whether there exists a significant difference in the mean age at first birth between urban and rural women in Jammu & Kashmir and Punjab.
- To analyze how education level and wealth status influences the age at first birth among women in both urban and rural areas.
- To compare overall patterns of age at first birth between Jammu & Kashmir and Punjab.

3.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

General Objective: To examine the socioeconomic and demographic factors associated with child survival among women aged 15-49 in Jammu & Kashmir and Punjab using NFHS-5 data.

Specific Objectives:

- To compare the magnitude and patterns of child survival between Jammu & Kashmir and Punjab in relation to their socioeconomic and demographic profiles.

- To assess how child survival in Jammu & Kashmir and Punjab is related to maternal socioeconomic status (wealth index and education level) and environmental and infrastructural conditions—specifically, type of residence (urban/rural), source of drinking water, and type of toilet facility.

3.5 Influence in employment of women

General Objective: To understand what are the socioeconomic and demographic factors that contribute to the employment of women in Jammu & Kashmir and Punjab using NFHS-5 data.

Special Objectives:

- To understand how much does the level of education, wealth index, type of residence (urban/rural) and state together influence the chances of employment of women.

- To understand what are the individual contributions of each of these factors.

4 Research hypothesis

4.1 Determinants of Fertility

- **Null Hypothesis (H_0):** Education, wealth, and urban-rural residence have no significant effect on the number of children.

- **Alternative Hypothesis (H_1):** At least one of these variables significantly affects the number of children.

4.2 Early Marriage in India: Prevalence and Predictors

- **H1:** Early marriage prevalence is higher in Punjab than in Jammu & Kashmir within each education stratum.

- **H2:** Higher educational attainment is associated with significantly lower odds of early marriage.

- **H3:** After adjustment, education will remain a stronger predictor of early marriage than wealth.

4.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

The following hypotheses were formulated to examine whether there exists any significant difference in the mean age at first birth between women residing in urban and rural areas across the two selected states. Specifically, the hypotheses are stated as follows:

- **Null hypothesis (H_0)**: The mean age at first birth among women in urban areas is equal to that of women in rural areas.
- **Alternative hypothesis (H_1)**: The mean age at first birth among women in urban areas is not equal to that of women in rural areas.

The null hypothesis makes no assumptions on wealth and education status.

4.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

We test the following null hypotheses against their respective alternate hypotheses (that the null hypothesis is false):

- There is no significant difference in under five child survival odds between Jammu & Kashmir and Punjab.
- Survival odds of children under five years of age in both states is not significantly influenced by maternal socioeconomic status (household wealth and education level) or by environmental and infrastructural factors (place of residence - urban/rural and access to improved water sources and sanitation facilities)

4.5 Influence on education

We want to test the following Hypothesis:

- **Null Hypothesis (H_0):** The variables - Education level, Wealth index, type of residence, state collectively do not influence the employment status of women in Punjab and Jammu & Kashmir.
  Alternative Hypothesis (H_1): The above variables collectively influence the employment status of women in Punjab and Jammu & Kashmir.

- For each of the variables - Education level, Wealth index, type of residence and state we test the following.

  Null Hypothesis (H_0): The factor individually does not influence the employment status of women in Punjab and Jammu & Kashmir.

  Alternate Hypothesis (H_1): The factor individually has significant influence on the employment status of women in Punjab.

5 Data sources

We have used data from the complete **NFHS 5 IR** for our analysis.

6 Methodology

6.1 Determinants of Fertility

To analyse the effect of mother's education on fertility behaviour, a survey-weighted quasi-Poisson regression model was employed. The dependent variable is the total number of children ever born to a woman ($V201$), while the key independent variables include: the mother's educational attainment ($V106$), household wealth index ($V190$), and the place of residence (urban/rural, $V025$).

The model is specified as follows:

$$\mathbb{E}(V201_i) = \exp(\beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Wealth}_i + \beta_3 \text{Urban/Rural}_i),$$

where:

- $\mathbb{E}(V201_i)$ represents the expected number of children ever born to the i^{th} woman,
- β_0 is the intercept term,
- Education_i denotes the woman's educational attainment level,
- Wealth_i represents the household wealth index,
- Urban/Rural_i indicates the place of residence (urban = 1, rural = 0),
- and β_1, β_2, β_3 are the respective regression coefficients.

Taking the logarithm of both sides gives the log-linear form:

$$\log(\mathbb{E}(V201_i)) = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Wealth}_i + \beta_3 \text{Urban/Rural}_i.$$

This specification ensures that the predicted number of children remains non-negative and allows for a multiplicative interpretation of the effects. Each $\exp(\beta_j)$ represents the rate ratio associated with a one-unit change in the corresponding predictor, holding other factors constant.

The quasi-Poisson approach was chosen because the dependent variable, *number of children ever born*, is a count variable that takes non-negative integer values. It also allows for overdispersion in the data (i.e., when the variance exceeds the mean), which is common in fertility data. The survey weight used in all analyses was the NFHS sampling weight `V005`, rescaled by dividing by 1,000,000, as recommended for DHS datasets.

6.2 Early Marriage in India: Prevalence and Predictors

6.2.1 Variable description

Outcome: `early_marriage` = 1 if age at first marriage < 18 (derived from the age-at-first-marriage variable), 0 otherwise.

Main predictors: Education: `V106` (categorical), used with levels: No education (0), Primary (1), Secondary (2), Higher (3). Wealth: collapsed into three groups `wealth3` = Poor (V190 = 1–2), Middle (V190=3), Rich (V190 = 4–5). Residence: Urban/Rural (V025).

Survey design: cluster = V001, strata = V022 (if present), weight = V005 scaled by 10^6.

6.2.2 Analytical strategy

1. Produce survey-weighted estimates of early marriage prevalence by state and education (`svyby(..., svymean)`).

2. For each education level, perform a survey-aware comparison of early-marriage prevalence between Punjab and J&K (survey-based two-sample test). Collect raw p-values and apply Benjamini–Hochberg (FDR) correction across the education-level tests.

3. Fit a multivariable survey-weighted logistic regression (`svyglm(..., family = quasibinomial)`) of `earl_marriage` on education (V106), wealth3, residence and state.

4. Present results as adjusted coefficients and odds ratios with 95% CIs. For presentation we rely on design-corrected standard errors (survey package).

6.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

The analysis was restricted to ever-married women aged 15–49 years from the states of Jammu & Kashmir and Punjab.

The dependent variable, **age at first birth**, was computed using the variable `V012`, which is precisely age of the respondent at first birth. The key explanatory variables include:

- **Type of residence** (`v025`): Urban / Rural
- **Educational attainment** (`v106`): No education, Primary, Secondary, Higher
- **Wealth index** (`v190`): Poorest, Poorer, Middle, Richer, Richest

Two main statistical techniques were applied:

1. **Two-sample *t*-test:** Conducted separately for Jammu & Kashmir and Punjab to test whether the mean age at first birth differs significantly between urban and rural women, assuming equal variances.
2. **Multiple Linear Regression:** Estimated using the Ordinary Least Squares (OLS) method to assess the joint effect of residence, education, and wealth on age at first birth:

$$\text{Age at First Birth} = \beta_0 + \beta_1(\text{Residence}) + \beta_2(\text{Education}) + \beta_3(\text{Wealth}) + \varepsilon$$

All analyses were carried out in `R` using the packages `dplyr`, `broom`, `survey` and `ggplot2`.

6.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

6.4.1 Dependent and Independent Variables Used

We list the variables which we have **constructed** from the NFHS 5 IR dataset for our analysis. The variables which have been used as-is are mentioned alongwith the models.

- `child-dead-under-5`: In order to construct this variable, we first reshape our data using `pivot_longer` command in `R`. This ensures the correctness and simplicity of further analysis by aiding in variable construction. The variable is constructed using the columns beginning with `B5` (`B5$i` records whether the i^{th} child of the respondent is dead or alive at the time of the interview) and `B7` (`B7$i` records the age at death in months if the i^{th} child of the respondent has died) in the NFHS dataset. We assign the variable to be 1 if the child is dead at the time of interview and their age at death is less than 60 months, and 0 otherwise (in cases where `B7` is `NA`, but `B5` is not `NA`, we assign the variable a value of 0). This variable acts as a proxy for measuring child survival under five years of age.

- `toilet`: This variable denotes the type of toilet facility available. It is constructed from the variable `V116` from the NFHS dataset, using one-hot encoding. It is a binary variable (1 = Improved, 0 = Unimproved), classified according to WHO JMP classification. (See Tables) We use Unimproved as our base reference.

- `water`: This is a binary variable (1 = Improved, 0 = Unimproved) constructed from the variable `V113`, using one-hot encoding, based on WHO JMP classification. (See Tables) We use Unimproved classification as our base reference.

- `V106`: This categorical variable encodes the level of education attained. It is already encoded in the dataset (0 = No education, 1 = Primary, 2 = Secondary, 3 =Higher) and used as-is. `No-education` classification is used as our base reference.

- `V190`: This categorical variable encodes the combined wealth index, encoded as (1 =Poorest, 2= Poorer, 3 = Middle, 4 = Richer, 5=Richest) in the dataset and used as-is. `Poorest` is our base reference.

- `V025`: This categorical variable encodes whether place of residence is urban or rural (1 = urban, 2 = rural) in the dataset and used as-is.

6.4.2 Differences in under-five mortality due to state and sex:

- A binary logistic regression model was used to examine sex differences in under-five child mortality across Punjab and Jammu & Kashmir (J&K).

- The dependent variable is `child-death-under5`, representing survival status for children upto five years of age($1 = \text{died before age five}, 0 = \text{survived beyond age five}$).

- The key independent variables are:
  - `B4` – Sex of the child ($1 = \text{male}, 2 = \text{female}$)
  - `state` – State indicator ($1 = \text{J\&K}, 3 = \text{Punjab}$)
  - `state:B4` – Interaction term to test whether the effect of sex on mortality differs by state
- The estimated model is given by:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{state}_i + \beta_2 \text{B4}_i + \beta_3(\text{state}_i \times \text{B4}_i)$$

  where p_i denotes the probability that the i^{th} child died before age five.

6.4.3 Determinants of under-five child survival:

- To assess the effect of socioeconomic and demographic factors on child survival, a multivariate logistic regression model was fitted.
- The dependent variable remains the same (`child-death-under5`).
- The explanatory variables included:
  - `V190` – Wealth index (categorical, proxy for household economic status)
  - `V106` – Mother's education level (Highest education level attained)
  - `V025` – Place of residence ($1 = \text{urban}, 2 = \text{rural}$)
  - `water` – Source of drinking water
  - `toilet` – Type of toilet facility (proxy for sanitation)
  - `state` – State indicator ($1 = \text{J\&K}, 3 = \text{Punjab}$)
- The model is specified as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{V190}_i + \beta_2 \text{V106}_i + \beta_3 \text{V025}_i + \beta_4 \text{water}_i + \beta_5 \text{toilet}_i + \beta_6 \text{state}_i + \varepsilon$$

- Multicollinearity among predictors was checked using the Variance Inflation Factor (VIF), and all variables were within acceptable limits ($\text{VIF} < 5$).

6.4.4 Survey design

Cluster = V001, Strata = V022 (if present), Weight = V005 scaled by 10^6.

6.5 Influence in employment of women

6.5.1 Variable Description

The two types of variables used in the inference were

Regressand/Outcome Variable -

- **V714** - Whether the respondent is currently working.

Regressor/Predictor Variable -

- **V106** - Education Level
- **V190** - Wealth index
- **V025** - Type of residence (Rural/ Urban)
- **V024** - State (Punjab / Jammu & Kashmir)

6.5.2 Survey design

Cluster = V001, Strata = V022 , Weight = V005 scaled by 10^6.

6.5.3 Methodology

- Produce survey-weighted estimates of employment status according to survey design using `survey` package.
- Fit a multivariable **survey-weighted Logistic Regression model** of the employment rate (regressand) on the predictors - Education level, Wealth index, Type of residence and State using `svyglm` function.
- To make inference about the collective influence of all the predictor variables, we conduct a **survey-weighted Wald test** on the model.
- To make inference about the individual influence of each of the predictor variables, we report the summary of the individual parameters of the model.

7 Salient findings

7.1 Determinants of Fertility

For the state of Jammu & Kashmir it was observed -

- **Education:** All education coefficients are negative and highly significant ($p < 0.001$). Compared to women with no education, those with primary, secondary, and higher education have progressively fewer children on average. Specifically, the estimated coefficients are:

$$\text{Primary: } -0.19, \quad \text{Secondary: } -0.87, \quad \text{Higher: } -1.63$$

  This indicates a strong inverse relationship between education and fertility: as education increases, the expected number of children decreases substantially.

- **Wealth:** The coefficients for the wealth index show mixed effects. The richest categories (particularly the fourth and fifth quintiles) have positive and statistically significant coefficients ($p < 0.001$), indicating that wealthier women tend to have slightly higher fertility compared to the poorest group, though the effect size is small.

- **Urban–Rural Residence:** The coefficient for rural residence is positive and significant ($\beta = 0.059$, $p = 0.002$), suggesting that women in rural areas tend to have marginally more children than those in urban areas.

- The model's dispersion parameter is 1.23, suggesting mild overdispersion handled appropriately by the quasi-Poisson specification.

- **Residuals vs Fitted Plot:** The residuals are centred around zero with no systematic pattern, indicating that the model captures the mean structure appropriately. The increasing spread of residuals at higher fitted values reflects overdispersion, supporting the use of a quasi-Poisson model.

- **Normal Q–Q Plot:** The deviance residuals deviate from the reference line, which is expected in count data models where normality is not assumed. However, the deviations are not severe, suggesting that the model assumptions are reasonably satisfied.

For the state of Punjab it was observed -

- The coefficients for `factor(V106)` (education level) are all negative and highly significant (`p < 0.001`), indicating a strong negative relationship between education and fertility.

- Women with primary, secondary, and higher education have progressively fewer children compared to uneducated women.

- Wealth index variables (`V190`) are not statistically significant, implying that once education is accounted for, wealth does not independently affect fertility behaviour.
- Residence type (`V025`) shows a weak negative association (`p = 0.051`), suggesting that urban women may have slightly fewer children than rural women, but this difference is only marginally significant.
- The dispersion parameter close to 1 (0.996) indicates a good model fit for the quasi-Poisson regression.
- **Residuals vs Fitted Plot:** The residuals are scattered widely without any clear pattern, indicating that the mean–variance relationship specified by the quasi-Poisson model is broadly appropriate. There is no strong curvature or systematic trend, suggesting that the model does not suffer from major misspecification in the mean structure.
- **Normal Q–Q Plot:** The deviance residuals deviate from the theoretical normal line, especially in the tails, showing heavier tail behaviour than expected under a normal distribution. This is typical for count data models and does not invalidate the quasi-Poisson approach. It mainly indicates that residuals are not normally distributed, which is not an assumption of the model.

7.2 Early Marriage in India: Prevalence and Predictors

7.2.1 Descriptive prevalence by education and state

Table 3 shows survey-weighted prevalence of early marriage by education category and state (95% CI from design-based variance).

7.2.2 Multiple-testing results (Benjamini–Hochberg)

We tested "Punjab vs J&K" difference in early marriage prevalence within each education level and adjusted p-values across tests using Benjamini–Hochberg (FDR). Results are shown in Table 4.

Interpretation: After controlling the false discovery rate, early marriage prevalence is significantly higher in Punjab than in J&K for Primary and Secondary educational groups with extremely small adjusted p-values, and also higher for the Higher educated group (adjusted p 0.0207). For the "No education" category the design-based test could not produce a valid p-value due to insufficient PSU/variance in that stratum.

7.2.3 Survey-weighted multivariable model

We fit a survey-weighted logistic regression (quasibinomial family) of early marriage on education (V106), collapsed wealth (wealth3), residence and state. The model accounts for the NFHS complex design (clusters, strata and sampling weights). A Wald test for the joint significance of the model terms (factor(V106), wealth3, residence, state) was highly significant (F = 144.53 on 7 and 1585 df, $p < 2.22 \times 10^{-16}$), indicating the set of predictors contributes to explaining early marriage.

Interpretation: After accounting for the survey design and adjusting for wealth and residence, educational attainment is strongly associated with reduced odds of early marriage. Compared to women with no education, those with *Primary* education have a modestly lower odds (OR ≈ 0.87, $p \approx 0.046$), those with *Secondary* education have substantially lower odds (OR ≈ 0.34, $p < 0.001$), and women with *Higher* education show a very large reduction in odds (OR ≈ 0.04, $p < 0.001$). Wealth shows a protective association only for the 'Rich' group (OR ≈ 0.84, $p \approx 0.006$), while the 'Middle' group is not significantly different from 'Poor'. Urban residence is associated with higher adjusted odds of early marriage (OR ≈ 1.24, $p < 0.001$). Finally, after adjustment, being in Punjab is associated with nearly double the odds of early marriage relative to Jammu & Kashmir (OR ≈ 1.99, $p < 0.001$).

7.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

A preliminary histogram of Age at First Birth was plotted for both states of Jammu and Kashmir and Punjab and across Urban and Rural areas.

The histograms illustrate the distribution of women's age at first birth across urban and rural areas in Jammu & Kashmir (J&K) and Punjab. In both states, the distribution is right-skewed, indicating that most women have their first child in their early twenties, with relatively fewer births occurring after age 30. In J&K, the urban distribution is shifted slightly to the right compared to the rural one, showing that urban women tend to delay childbirth. A similar pattern is visible in Punjab, though the difference between urban and rural areas is comparatively smaller. Overall, the mean age at first birth is slightly higher in J&K than in Punjab, suggesting later childbearing patterns in J&K. Across both states, rural women exhibit earlier childbearing, reflecting differences in socioeconomic conditions.

Jammu & Kashmir: Urban–Rural Comparison

A two-sample t-test assuming equal variances was conducted to compare the mean age at first birth between urban and rural women in Jammu & Kashmir. The results indicate a statistically significant difference in the mean ages ($t = 10.571$, $df = 13178$, $p < 0.001$). The mean age at first birth among urban women (23.92 years) is higher than that of rural women (22.89 years), with a 95% confidence interval for the difference ranging from 0.83 to 1.21 years. This suggests that urban women tend to have their first birth at a relatively later age compared to rural women.

Punjab: Urban–Rural Comparison

A two-sample t-test assuming equal variances was performed to compare the mean age at first birth between urban and rural women in Punjab. The test revealed a statistically significant difference between the two groups ($t = 5.139$, $df = 14817$, $p < 0.001$). The mean age at first birth among urban women (22.30 years) is slightly higher than that of rural women (21.94 years), with a 95% confidence interval for the difference ranging from 0.22 to 0.49 years. This indicates that women residing in urban areas tend to have their first birth at a marginally later age compared to their rural counterparts.

Jammu & Kashmir: Regression Based Analysis of Determinants of Age at First Birth

The multiple linear regression results for Jammu & Kashmir indicate that type of residence, educational attainment, and household wealth collectively exert a statistically significant influence on the mean age at first birth ($F(8, 13171) = 99.91$, $p < 0.001$).

Urban women tend to have a higher mean age at first birth than their rural counterparts, with residence showing a negative and significant coefficient for rural areas ($\beta = -0.35$, $p < 0.01$). Education exhibits a strong positive association with age at first birth: women with primary, secondary, and higher education have progressively higher mean ages at first birth compared to those with no education, with the largest increase observed among those with higher education ($\beta = 3.04$, $p < 0.001$). Wealth also shows a positive gradient, with middle, richer, and richest groups having significantly higher ages at first birth relative to the poorest, though the effect size is smaller than that of education. Overall, higher socioeconomic status and education are associated with delayed childbearing in Jammu & Kashmir.

Punjab: Regression Based Analysis of Determinants of Age at First Birth

In Punjab, the model also reveals statistically significant effects of residence, education, and wealth on age at first birth ($p < 0.001$). Urban residence is associated with a higher mean age at first birth compared to rural areas ($\beta = 0.26$, $p < 0.001$). The impact of education is again pronounced: women with secondary and higher education have significantly higher ages at first birth, particularly those with higher education ($\beta = 4.79$, $p < 0.001$). However, the primary education category does not show a significant difference from women with no education.

7.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

1. **Under-Five Deaths by State and Sex**

    - The weighted analysis reveals that **Punjab** exhibits a substantially higher rate of under-five deaths compared to **Jammu & Kashmir (J&K)**.
    - Male children in Jammu & Kashmir show significantly **higher mortality odds** than females, whereas in Punjab the mortality odds of females is similar to that of males.
    - Logistic regression confirms this pattern:
        - The coefficient for *state* (Punjab) is positive and statistically significant, indicating an elevated risk of child death under five in Punjab.
        - The interaction term between *state* and *sex of the child* suggests that the sex-specific mortality gap differs by state, though the effect is not statistically significant at conventional levels of 1% or 5%.

2. **Determinants of Child Survival**

    - The multivariate logistic regression highlights the following associations:
        - **Wealth (V190)**: Negatively associated with mortality ($OR = 0.88$); children from wealthier households are less likely to die before age five.
        - **Education (V106)**: Strongest protective effect ($OR \approx 0.74$), suggesting that maternal education improves child survival outcomes.
        - **Residence (V025)**: Type of residence (urban/rural) does not have a significant effect on Child Survival Under Five years of age.
        - **toilet** and **water**: Show limited direct effects after accounting for socioeconomic variables, implying potential mediation through wealth or education.
        - **State (Punjab)** remains a strong and significant predictor ($OR = 3.49$), highlighting persistent regional disparities beyond household characteristics.

- **Multicollinearity Assessment**
  - Variance Inflation Factors (VIFs) for all predictors were below 2, indicating no serious multicollinearity.
  - However, sanitation and source of drinking-water were moderately correlated with wealth and education, which may partially explain their weaker independent effects.

7.5 Influence on employment of Women

1. **Collective Influence of regressor variables -** The **survey-weighted Wald F-statistic** is **F=5.09**, which, in our case has **degrees of freedom** $(4, 367)$. The computed **p-value is 0.00053**, which is very less, $p < 0.001$. This means that the regressor variables - Education, Wealth Index, Type of settlement, State together have a highly significant effect on the outcome variable - Employment status of women in the states of Punjab and Jammu&Kashmir. This shows enough evidence to reject our null hypothesis that these variables do not influence the Employment Status of women. The next step is to look for the effects of the individual regressors.

2. **Individual Influence of regressor variables -** The summary of the fit of the **survey-weighted Logistic Regression** model gives the following inference about the individual variables.

   - **Level of Education** - The p-value for the education variable (**V106**) is 0.09984, which is not very significant. Thus, from the model, level of education does not significantly influence the employment status of women in the states of Punjab and Jammu & Kashmir.
   - **Wealth Index** - The p-value for the wealth variable (**V190**) is 0.01399, which is significant, and the value of the coefficient is -0.10368, which is negative. Thus, from the model, wealth of an individual significantly influences the employment status of women, and wealthy individuals are less likely to be working.
   - **Type of Settlement** - The p-value for the settlement variable (**V025**) is 0.00176, which is highly significant. Thus, from the model, type of settlement has a highly significant influence on the employment status of women in the states of Punjab and Jammu & Kashmir.
   - **State** - The p-value for the state variable (**V024**) is 0.64118, which is not at all significant. Thus, from the model, the state does not significantly influence the employment status of women in this context. So, if the other factors are same, then whether an individual is from Punjab or she is from Jammu & Kashmir will not have significant influence on determining

her employment status. For this reason, we do not conduct any separate state-wise test for the other factors.

8 Discussion

8.1 Determinants of Fertility

The findings reveal a clear and statistically robust inverse relationship between women's education and fertility in both the states. Each successive level of education is associated with a substantial decline in the number of children ever born, highlighting the critical role of female education in fertility reduction.

Wealth effects are comparatively weaker, suggesting that economic status alone does not strongly predict fertility once education is accounted for. The slight fertility difference between rural and urban women may reflect varying access to family planning and cultural norms regarding family size.

Overall, the results underscore that expanding female education—particularly secondary and higher education—can be an effective policy measure to promote lower fertility and empower women in both the state.

8.2 Early Marriage in India: Prevalence and Predictors

- **Novelty:** We combined survey-corrected comparisons across education strata with a multiple-testing correction (Benjamini–Hochberg) to robustly test the "Punjab vs J&K" hypothesis.
- **Key messages:** Education is strongly protective against early marriage. Punjab shows higher prevalence than J&K even after adjustments. Wealth (collapsed) was not significant in the adjusted model, suggesting that education may be a stronger structural determinant in this context.
- **Limitations:** Subsetting to states reduces PSU/stratum counts and in some narrow strata (e.g., no-education) design-based tests can be undefined (NA). The analysis is cross-sectional; causal language should be avoided.

8.3 Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

This study highlights the persistent urban–rural gap in the age at first birth across both Jammu & Kashmir and Punjab, with urban women tending to have their first child later than their rural counterparts. Education and wealth emerge as strong predictors of delayed childbearing, underscoring the influence of socio-economic empowerment on reproductive behavior. The findings emphasize the need for policies promoting female education and economic inclusion to address early childbearing, particularly in rural settings.

Further exploration can include a multilevel regression approach to account for clustering at district or household levels, as the NFHS data follow a hierarchical sampling design. Inclusion of additional covariates such as employment status, age at marriage, and contraceptive use can provide a more comprehensive understanding of the determinants of age at first birth. Temporal comparisons using previous NFHS rounds may also help in identifying trends and changes over time. Moreover, interaction effects between education, wealth, and residence can be analyzed to uncover complex socio-economic dynamics influencing reproductive behavior.

8.4 Child Survival Under Five - Determinants and Patterns across Jammu&Kashmir and Punjab

The analysis of under-five child mortality in Punjab and Jammu & Kashmir (J&K) using the NFHS-5 dataset highlights important regional and socioeconomic disparities in child survival outcomes. The results indicate that Punjab exhibits a significantly higher rate of under-five mortality compared to J&K.

Sex-based differences in child survival reveal an intriguing pattern. In Jammu and Kashmir, male children are more likely to die before age five than female children, while in Punjab there is no significant difference in the mortality odds of male and female children below five years of age.

Among the socioeconomic determinants, wealth and maternal education emerge as the most consistent protective factors. Children born to wealthier and more educated mothers face markedly lower mortality risks, consistent with established evidence linking socioeconomic status to child health outcomes. Access to improved sanitation and drinking water shows weaker effects after accounting for household wealth, indicating that these infrastructure variables may operate through economic and educational pathways rather than exerting direct independent influence.

The multicollinearity diagnostics further support the robustness of these findings. Variance Inflation Factors (VIFs) below 2 indicate minimal redundancy among predictors, strengthening confidence in the model estimates. However, the moderate correlations between sanitation, water, and wealth suggest overlapping mechanisms that warrant further exploration through path or mediation analysis.

Overall, these findings underscore the need for region-specific child health interventions. In Punjab, efforts may need to focus on addressing health service access, while in Jammu&Kashmir continued investment in maternal education and household welfare could sustain already favourable child survival trends. The persistence of significant state-level effects, even after controlling for individual and household factors, highlights the importance of considering contextual determinants—such as governance quality, healthcare delivery systems, and sociocultural norms—in reducing child mortality disparities across regions.

8.5 Influence on employment of women

We have observed from our inference that type of settlement and wealth are dominant influencers of working status of women in the states of Punjab and Jammu & Kashmir, while education is not a dominant influencer of it. This suggests that the main occupation of the people in those areas come from activities that do not require technical knowledge. Also, urban people are in general more likely to be employed, which seems reasonable since they have more exposure to opportunities and facilities for work. In our inference we observed that more wealth has association to lower work in those areas, which suggests that in those regions women from wealthy families do not experience the necessity to work, or, due to social taboos, are discouraged from working.

9 Conclusion

This study highlights that despite geographic proximity, Punjab and Jammu & Kashmir differ significantly in multiple demographic indicators, and these differences are strongly stratified by socioeconomic characteristics. Education consistently emerged as a key determinant across most domains—lower fertility, lower likelihood of early marriage, delayed age at first birth, improved child survival, and higher probability of women's employment. Wealth and place of residence also played important roles, although their effects were more domain-specific. For instance, rural disadvantage remained visible in fertility and age at first birth, whereas state-level contrasts were particularly pronounced in early marriage and women's employment patterns. Taken together, the findings reinforce that demographic outcomes cannot be understood purely as state-level aggregates but must be interpreted jointly with structural inequities and contextual social determinants. Policies aimed at improving women's education and enhancing economic opportunities are likely to have wide-ranging demographic benefits. Moreover, targeted and state-specific approaches may be needed, since the direction and magnitude of associations vary between Punjab and Jammu & Kashmir. This comparative evidence therefore provides a data-driven basis for evolving more regionally sensitive interventions that strengthen women's well-being and improve long-term population health outcomes in India.

References

- Vikram K, Vanneman R. *Maternal education and the multidimensionality of child health outcomes in India*. *J Biosoc Sci.* 2020 Jan;52(1):57-77. doi: 10.1017/S0021932019000245. Epub 2019 May 21. PMID: 31112112; PMCID: PMC7068132.
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Tables

Determinants of Fertility

Table 1: Survey-weighted quasi-Poisson regression results for Jammu & Kashmir

Variable	**Estimate**	**Std. Error**	**t value**	**Pr(> $\|t\|$)**
Intercept	0.8037	0.0265	30.354	<2e-16***
Education (Primary)	-0.1923	0.0241	-7.993	4.91e-15***
Education (Secondary)	-0.8724	0.0159	-54.818	<2e-16***
Education (Higher)	-1.6275	0.0370	-44.003	<2e-16***
Wealth (Poorer)	-0.0078	0.0230	-0.338	0.735
Wealth (Middle)	0.0245	0.0251	0.979	0.328
Wealth (Richer)	0.1105	0.0247	4.478	8.68e-06***
Wealth (Richest)	0.2890	0.0273	10.600	<2e-16***
Residence (Urban)	0.0587	0.0192	3.058	0.0023**
Dispersion parameter		1.234		
Fisher Scoring iterations		6		

Signif. codes: *** 0.001; ** 0.01; * 0.05

Table 2: Survey-weighted quasi-Poisson regression results for Punjab

Variable	**Estimate**	**Std. Error**	**t value**	**Pr(> $\|t\|$)**
Intercept	1.0317	0.0623	16.560	<2e-16***
Education (Primary)	-0.2062	0.0205	-10.036	<2e-16***
Education (Secondary)	-0.8156	0.0170	-47.935	<2e-16***
Education (Higher)	-1.3622	0.0276	-49.274	<2e-16***
Wealth (Poorer)	-0.0494	0.0671	-0.736	0.462
Wealth (Middle)	-0.0325	0.0629	-0.516	0.606
Wealth (Richer)	0.0077	0.0619	0.125	0.901
Wealth (Richest)	0.0957	0.0629	1.523	0.128
Residence (Urban)	-0.0282	0.0144	-1.953	0.051·
Dispersion parameter		0.996		
Fisher Scoring iterations		5		

Signif. codes: *** 0.001; ** 0.01; * 0.05; · 0.1

Table 3: Survey-weighted prevalence of early marriage (age at first marriage <18) by education and state

State	Education level	Prevalence	95% CI
J&K	No education	0.2094	(0.1968, 0.2219)
Punjab	No education	0.4084	(0.3904, 0.4264)
J&K	Primary	0.1733	(0.1519, 0.1948)
Punjab	Primary	0.3320	(0.3117, 0.3523)
J&K	Secondary	0.0688	(0.0639, 0.0738)
Punjab	Secondary	0.1106	(0.1047, 0.1165)
J&K	Higher	0.0131	(0.0094, 0.0168)
Punjab	Higher	0.0078	(0.0052, 0.0104)

Early Marriage in India: Prevalence and Predictors

Table 4: Raw and FDR-adjusted p-values for Punjab > J&K early-marriage prevalence by education

Education	Raw p-value	BH-adjusted p-value
No education	NA	NA
Primary	3.18×10^{-21}	4.77×10^{-21}
Secondary	5.76×10^{-25}	1.73×10^{-24}
Higher	0.0207	0.0207

Table 5: Survey-weighted logistic regression coefficients (quasibinomial)

Term	Estimate	Std. Error	t value	Pr(>—t—)
(Intercept)	-1.02783	0.05002	-20.547	$< 2 \times 10^{-16}$
Primary (V106 = 1)	-0.13736	0.06866	-2.000	0.04563 *
Secondary (V106 = 2)	-1.06558	0.05329	-19.995	$< 2 \times 10^{-16}$
Higher (V106 = 3)	-3.28995	0.16657	-19.751	$< 2 \times 10^{-16}$
wealth3 (Middle)	0.02653	0.06848	0.387	0.69847
wealth3 (Rich)	-0.17971	0.06524	-2.755	0.00594 **
Urban residence	0.21336	0.05419	3.937	8.6×10^{-5} ***
Punjab state	0.68591	0.04935	13.899	$< 2 \times 10^{-16}$

Table 6: Odds ratios (exponentiated coefficients)

Term	Odds Ratio (exp(coef))
Intercept	0.3578
Primary (vs No education)	0.8717
Secondary (vs No education)	0.3445
Higher (vs No education)	0.0373
Wealth3 Middle (vs Poor)	1.0269
Wealth3 Rich (vs Poor)	0.8355
Urban residence (vs Rural)	1.2378
Punjab (vs J&K)	1.9856

Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

Table 7: Mean Age at First Birth by Residence in J&K and Punjab

Residence	Mean Age (J&K)	Mean Age (Punjab)
Urban	23.92	22.30
Rural	22.89	21.94

Table 8: Two Sample t-test Results for J&K

data:	age_first_birth by residence
t =	10.571
df =	13178
p-value	< 2.2e-16
alternative hypothesis:	true difference in means between group Urban and group Rural is not equal to 0
95 percent confidence interval:	0.8320607 1.2108637
sample estimates:	
mean in group Urban	23.91537
mean in group Rural	22.89391

Table 9: Two Sample t-test Results for Punjab

data:	age_first_birth by residence
t =	5.139
df =	14817
p-value	= 2.797e-07
alternative hypothesis:	true difference in means between group Urban and group Rural is not equal to 0
95 percent confidence interval:	0.2213868 0.4944059
sample estimates:	
mean in group Urban	22.29709
mean in group Rural	21.93920

Determinants and Patterns of Child Survival Under Five Years of Age

The Improved and Unimproved water sources and sanitation facilities are given in Tables 10 and 11 respectively.

Table 10: Classification of Drinking Water Sources (Variable: V113)

Code	**Category (Label)**	**JMP Classification**
11	Piped into dwelling	Improved
12	Piped to yard/plot	Improved
13	Public tap / standpipe	Improved
21	Tube well / borehole	Improved
31	Protected dug well	Improved
32	Protected spring	Improved
33	Rainwater	Improved
41	Unprotected dug well	Unimproved
42	Unprotected spring	Unimproved
43	Surface water (river, dam, lake, pond, stream, canal, irrigation channel)	Unimproved
51	Tanker truck	Unimproved
52	Cart with small tank / drum	Unimproved
61	Bottled water	Improved
96	Other	Unimproved
99	Missing	–

Table 11: Classification of Sanitation Facilities (Variable: V116)

Code	Category (Label)	JMP Classification
11	Flush to piped sewer system	Improved
12	Flush to septic tank	Improved
13	Flush to pit latrine	Improved
14	Flush to somewhere else	Unimproved
15	Flush, don't know where	Unimproved
21	Ventilated improved pit (VIP) latrine	Improved
22	Pit latrine with slab	Improved
23	Pit latrine without slab / open pit	Unimproved
31	Composting toilet	Improved
41	Hanging toilet / latrine	Unimproved
42	Bucket toilet	Unimproved
95	No facility / bush / field	Unimproved
96	Other	Unimproved
99	Missing	–

The logistic regression results are as given in Tables 12 and 13.

Table 12: Survey-weighted logistic regression results of under-five mortality by state and child's sex

	Dependent variable:
	Child Died Under 5 (1 = Yes, 0 = No)
State	0.387^{***}
	(0.098)
Sex of Child	-0.297^{*}
	(0.166)
State × Sex of Child	0.105^{*}
	(0.063)
Constant	-4.297^{***}
	(0.257)
Observations	64,647
Note:	*p<0.1; **p<0.05; ***p<0.01
	Logistic regression with standard errors in parentheses.

Table 13: Survey-weighted multivariate logistic regression of under-five mortality on socioeconomic and infrastructural factors

	Dependent variable:
	Child Died Under 5 (1 = Yes, 0 = No)
Wealth Index	-0.120^{***}
	(0.037)
Maternal Education	-0.306^{***}
	(0.038)
Rural	-0.057
	(0.085)
Improved Sanitation	0.002
	(0.002)
Improved Drinking Water	-0.001
	(0.003)
State	1.251^{***}
	(0.087)
Constant	-3.415^{***}
	(0.229)
Observations	64,647
Note:	*p<0.1; **p<0.05; ***p<0.01
	Logistic regression with standard errors in parentheses.

Influence on employment of women

Table 14: Wald Test for Factors of Women's Employment Status

Model:	`svyglm(formula = V714 ~ V106 + V190 + V025 + V024, design = des, family = quasibinomial())`
Variables Tested:	V106, V190, V025, V024
F-statistic:	5.085978
Degrees of Freedom:	4 and 367
p-value:	0.00053373

Note: Significant at 0.001 level. Indicates that at least one *predictor variable significantly influences women's employment status.*

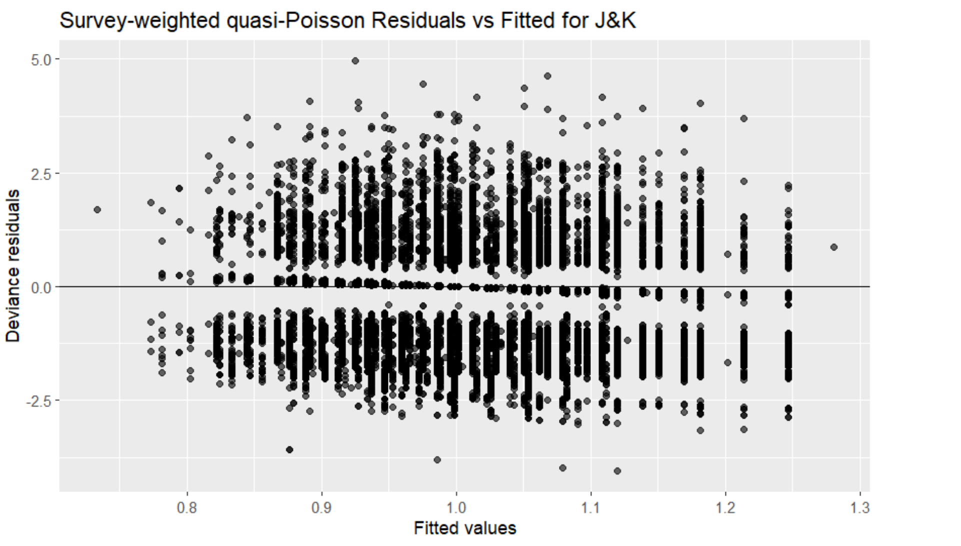
Table 15: Survey-weighted logistic regression results for determinants of women's employment status

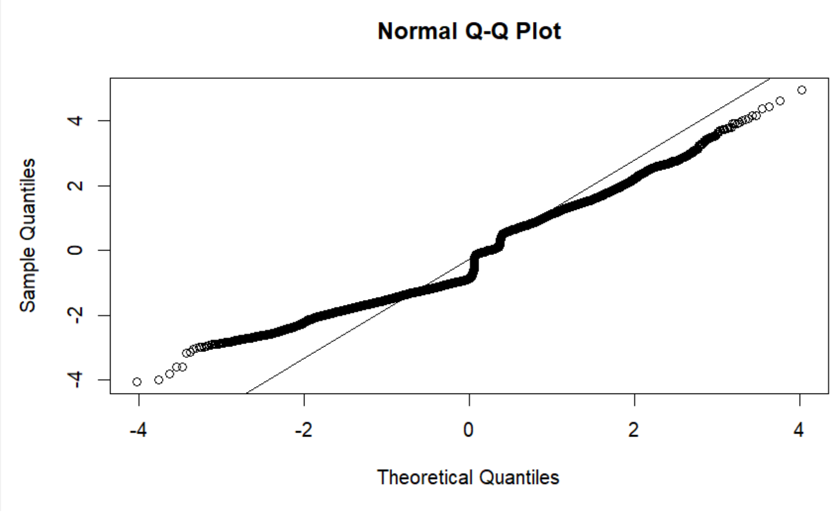
Variable	**Estimate**	**Std. Error**	**t value**	**Pr**$(>\|t\|)$
(Intercept)	-0.28543	0.24443	-1.168	0.24366
V106 (Education)	-0.07670	0.04649	-1.650	0.09984·
V190 (Wealth Index)	-0.10368	0.04198	-2.469	0.01399*
V025 (Residence)	-0.30521	0.09684	-3.152	0.00176**
V024 (State)	0.02642	0.05664	0.466	0.64118

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '·' 0.1 ' ' 1
Dispersion parameter for quasibinomial family taken to be 1.038606
Number of Fisher Scoring iterations: 4

Figures

Determinants of Fertility




Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
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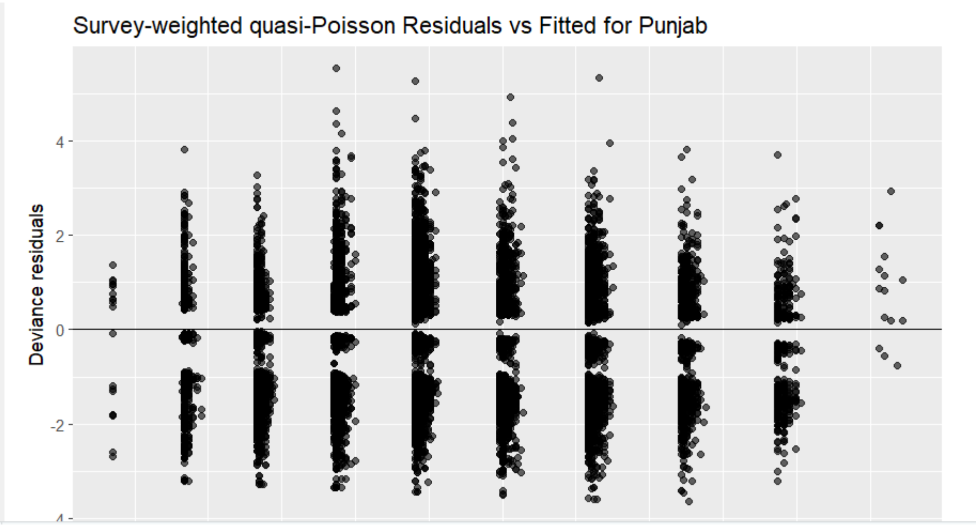
-2
0
2
4


Survey-weighted quasi-Poisson Residuals vs Fitted for Punjab
Deviance residuals
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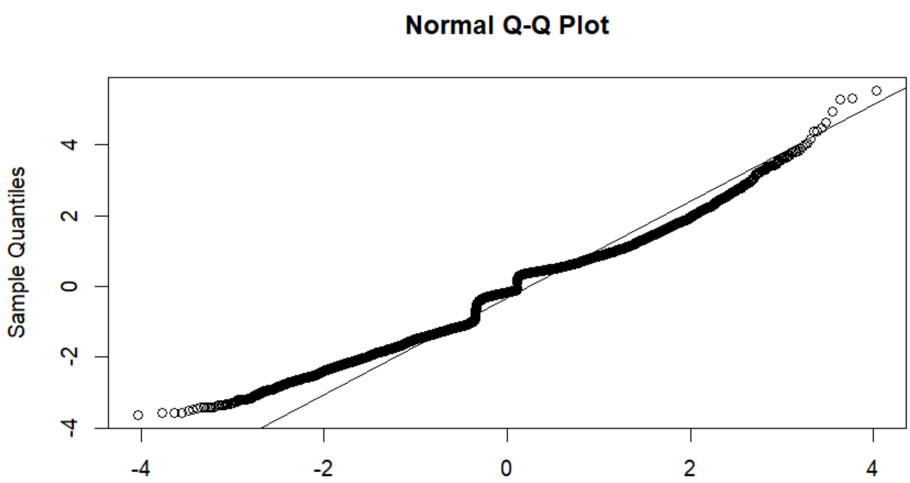
0
-2


Normal Q-Q Plot
Sample Quantiles
-4
-2
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2
4

Early Marriage in India: Prevalence and Predictors

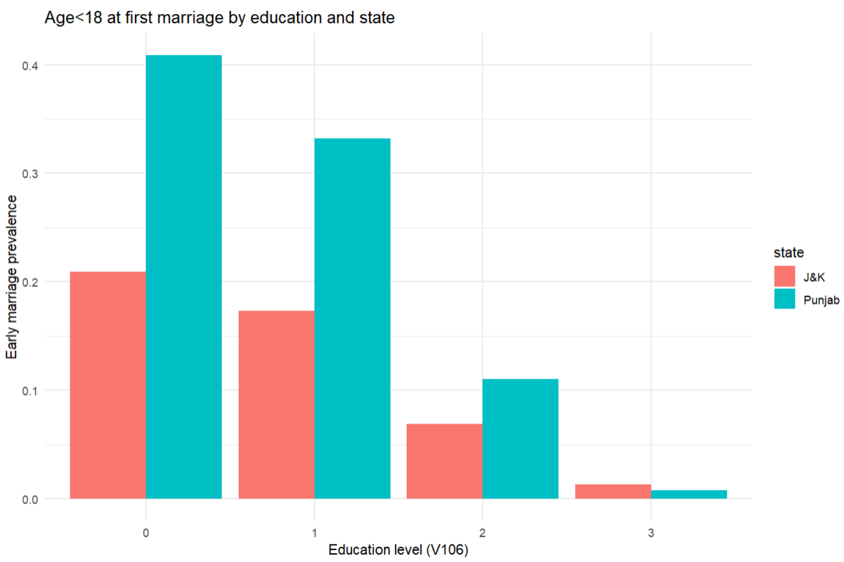

Figure 1: Survey-weighted prevalence of early marriage by education level and state

Determinants and Patterns of Child Survival Under Five Years of Age

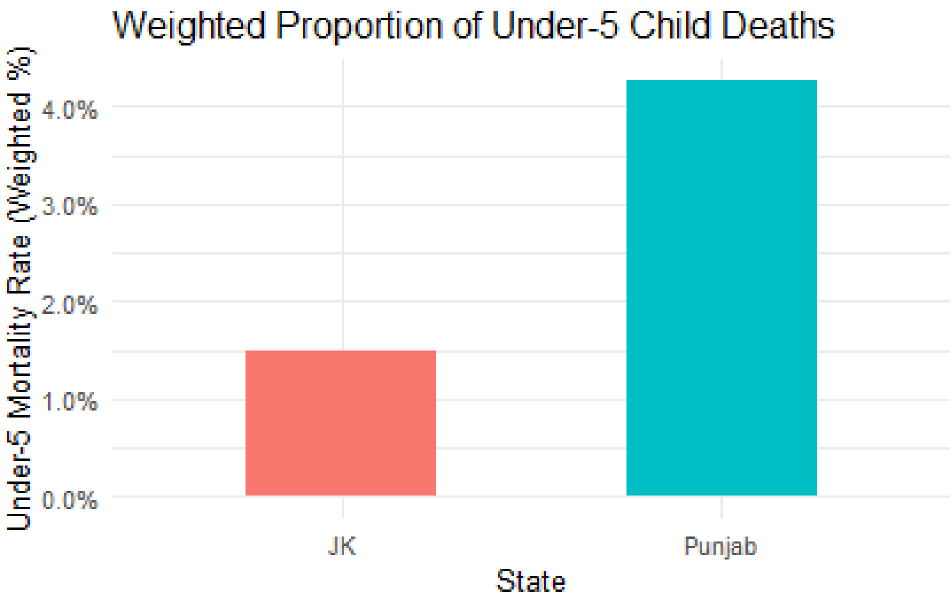

Figure 2: Weighted Proportion of Under Five Deaths across Punjab and Jammu&Kashmir

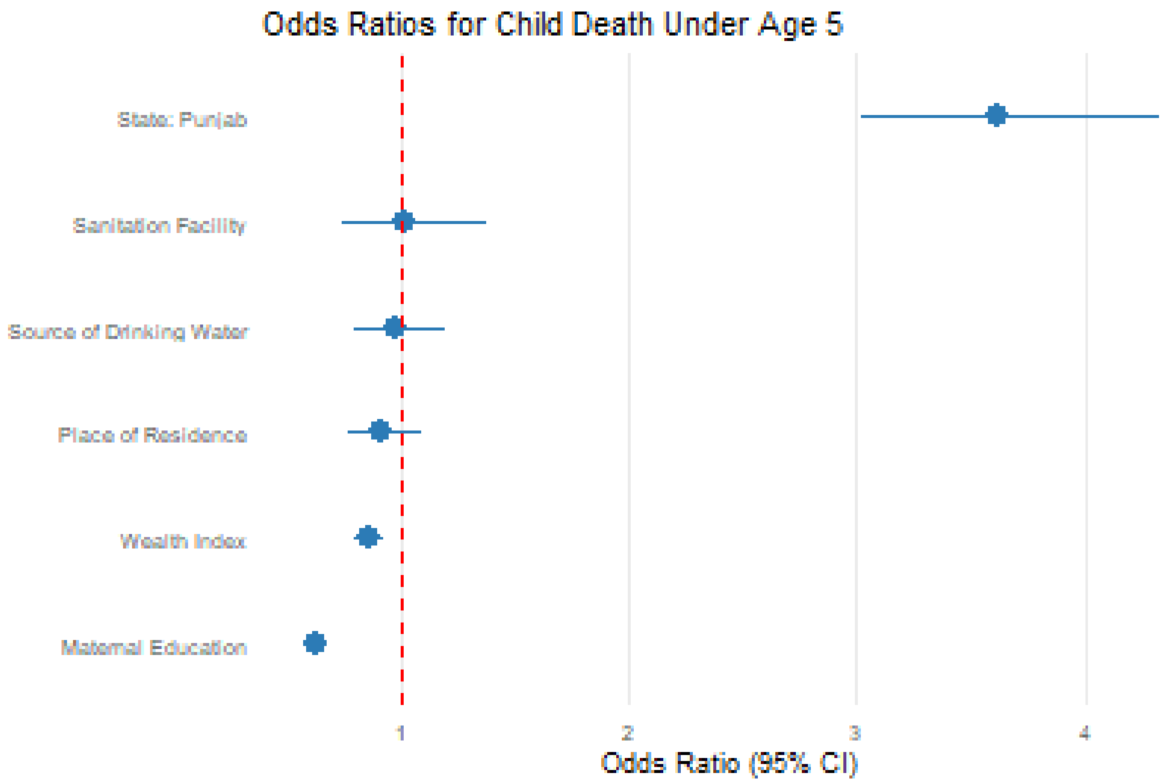

Figure 3: Determinants of Under Five Child Deaths across Punjab and Jammu & Kashmir

Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

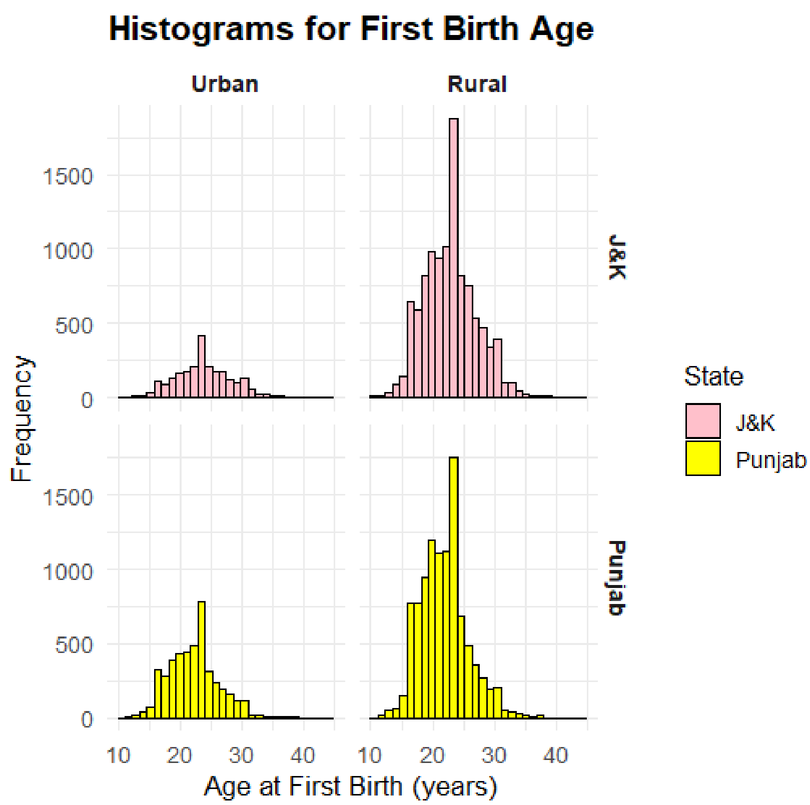

Figure 4: A histogram of age at first birth among urban and rural women in Jammu & Kashmir and Punjab.

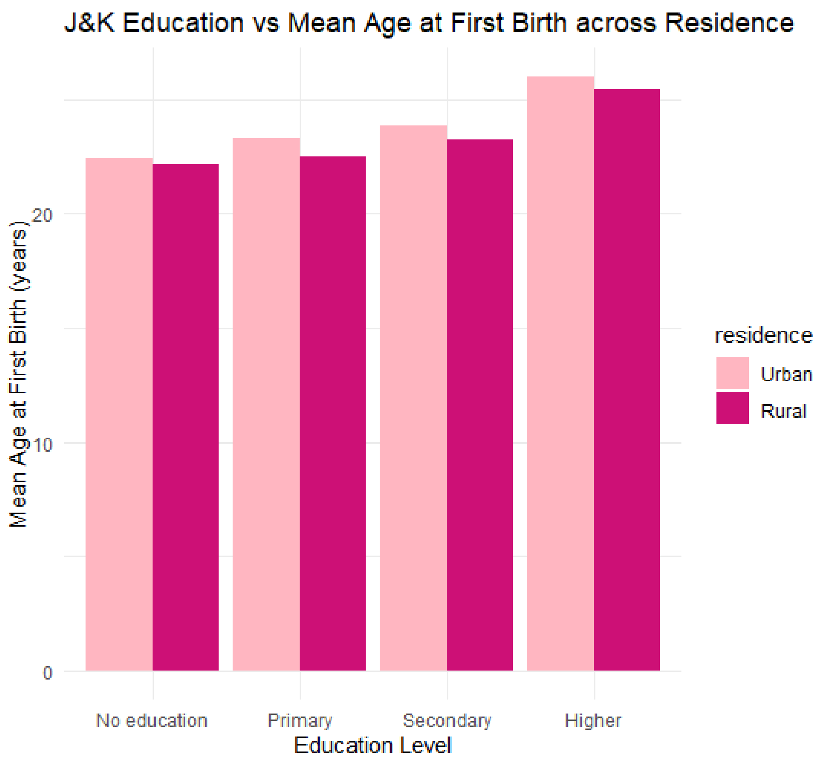

Figure 5: Barplot of Education vs Mean age at first birth in Jammu and Kashmir

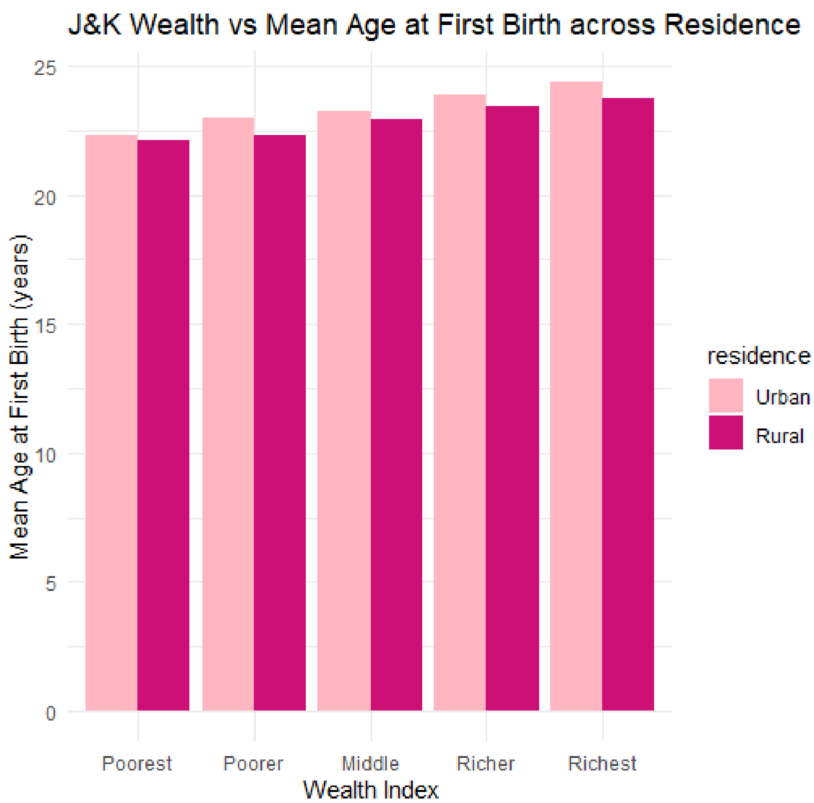

Figure 6: Barplot of Wealth vs Mean age at first birth in Jammu and Kashmir

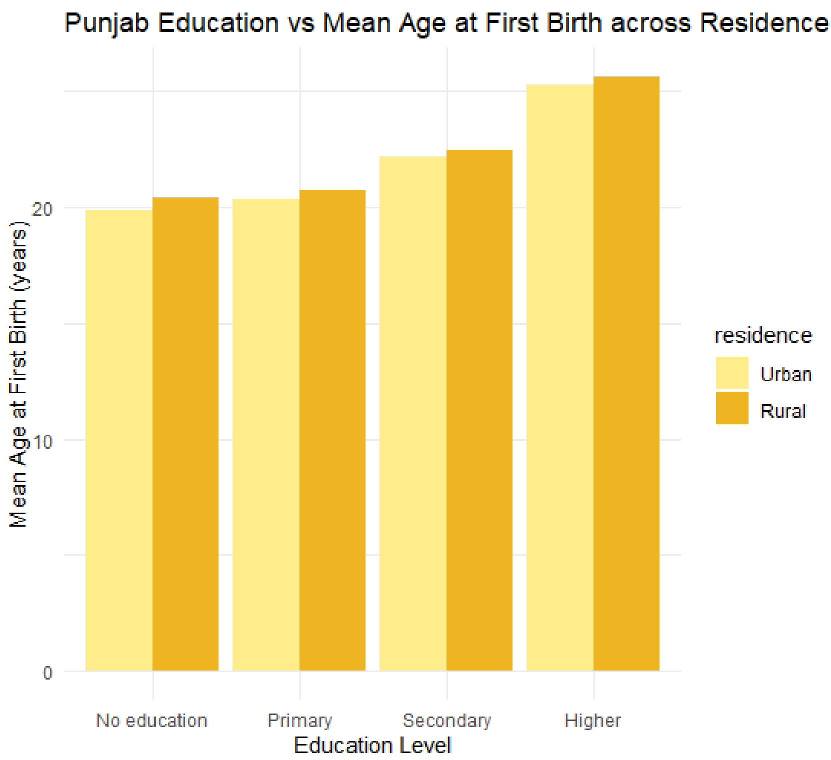

Figure 7: Barplot of Education vs Mean age at first birth in Punjab

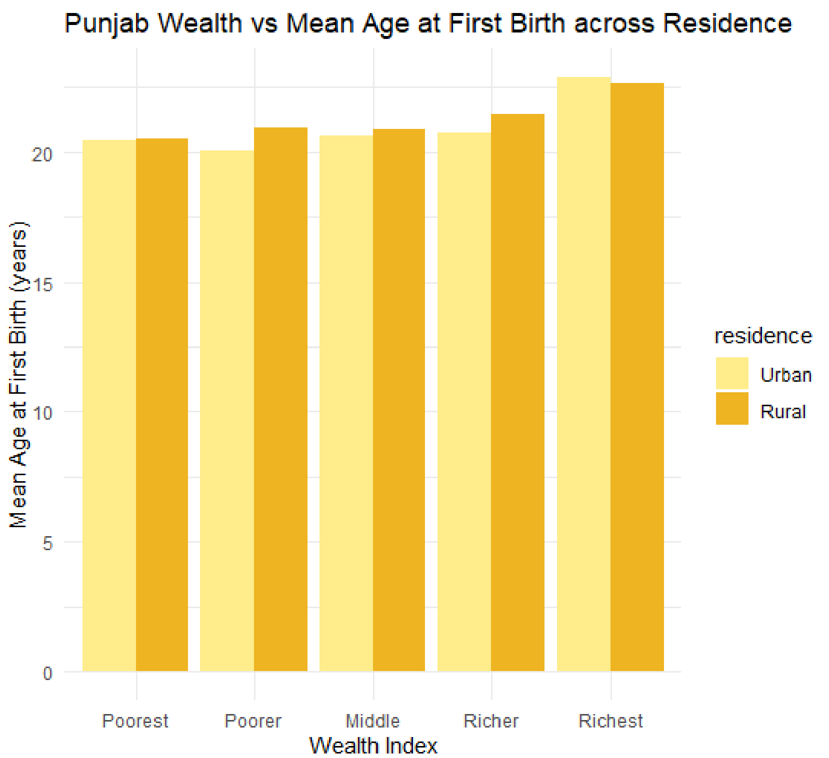

Figure 8: Barplot of Wealth vs Mean age at first birth in Punjab

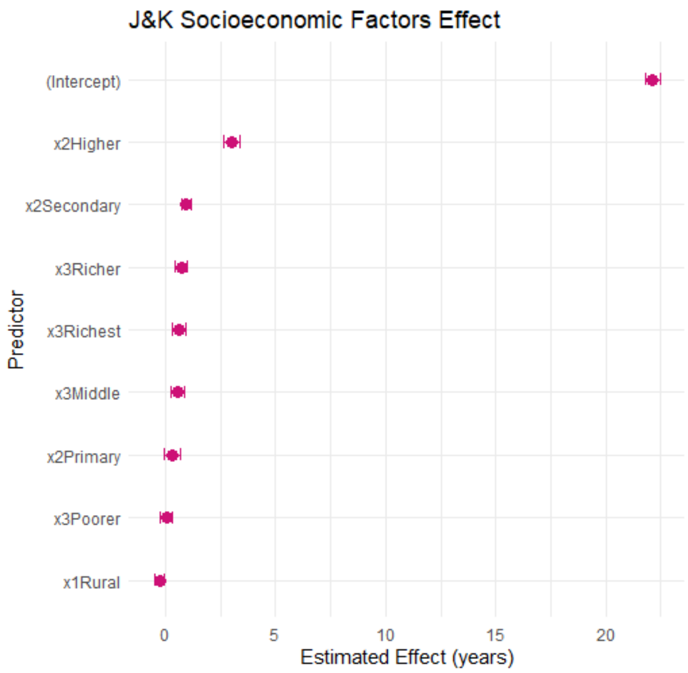

Figure 9: Plot showing the effect of covariates in regression for J&K

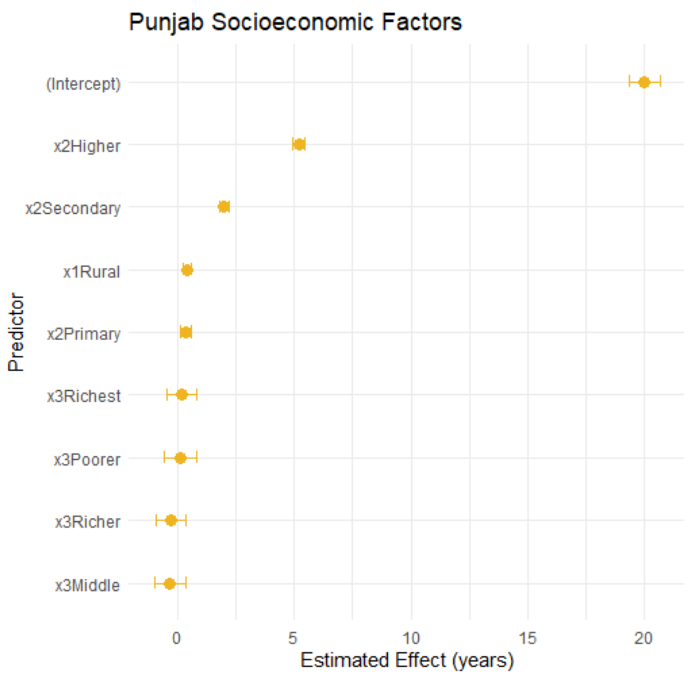

Figure 10: Plot showing the effect of covariates in regression for Punjab

Appendix

Determinants of fertility

Coding of Variables

The regression model employs the following key variables drawn from the NFHS dataset:

- **Dependent variable ($V201$):** Total number of children ever born to a woman.
- **Education level ($V106$):** Categorical variable representing educational attainment, recoded as:

No education, Primary, Secondary, Higher.

- **Wealth index ($V190$):** Categorical variable representing the household's wealth status
- **Place of residence ($V025$):** denoting the type of residence:

Interpretation of Coefficients

For categorical variables, the first category is taken as the reference group. Hence, the reported coefficients represent the difference in the expected log count of children compared to this baseline.

For instance, in the case of the residence variable ($V025$), "Urban" serves as the reference group. A positive and significant coefficient for "Residence (Rural)" indicates that, holding other variables constant, women living in rural areas are expected to have more children than those residing in urban areas.

The quasi-Poisson model estimates the relationship on the log scale:

$$\mathbb{E}(V201_i) = \exp(\beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Wealth}_i + \beta_3 \text{Urban/Rural}_i),$$

so a coefficient $\beta_3 = 0.059$ implies that rural women, on average, have $\exp(0.059) - 1 \approx 0.06$ or roughly 6% more children than their urban counterparts.

Early Marriage in India: Prevalence and Predictors

Data and variables (concise)

Data source: NFHS-5 (Individual Recode, IR) subset for Punjab and Jammu & Kashmir. Key variables used in the analysis:

- Outcome: `early_marriage` = indicator for age at first marriage < 18 years.
- Education: `V106` (categorical): No education (0), Primary (1), Secondary (2), Higher (3).
- Wealth: collapsed into `wealth3` with levels Poor (V190 = 1–2), Middle (V190 = 3), Rich (V190 = 4–5).
- Residence: Urban/Rural (V025).
- Survey design: cluster = V001, strata = V022, sampling weight = V005 (scaled by 10^6 in analysis).

Model specification (compact)

We estimated a design-corrected logistic regression using the `svyglm(..., family = quasibinomial)` framework:

$$\text{logit}\big[P(\text{early\_marriage} = 1)\big] = \beta_0 + \beta_1\text{V106} + \beta_2\text{wealth3} + \beta_3\text{residence} + \beta_4\text{state}.$$

All results reported account for clustering, stratification and sampling weights (Taylor linearization variance).

Key model diagnostics

Joint (Wald) test for the predictors (education, wealth3, residence, state): $F = 144.53$ on 7 and 1585 degrees of freedom, $p < 2.22 \times 10^{-16}$. This indicates the covariates jointly provide highly significant explanatory power for early marriage.

Notes on design-related caveats

- All inference above is design-based. Standard errors use the survey's clusters (V001) and strata (V022) and the sampling weight V005 scaled by 10^6.
- In small subgroups (e.g., some single education×state cells), variance estimation may be unstable or not computable; that is the reason some per-education tests returned NA for the p-value. We reported and explained this in the Results.

Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth

Statistical Tests Conducted

The analysis employed both inferential and regression-based methods to study urban–rural and socioeconomic differentials in age at first birth across Jammu & Kashmir and Punjab. The main techniques used were:

1. **Independent Two-Sample t-test:** Conducted to examine whether the mean age at first birth differs significantly between urban and rural women within each state.

   - For **Jammu & Kashmir**, the test indicated a significant difference ($t = 10.57$, p ¡ 0.001), with urban women having a higher mean age (23.92 years) compared to rural women (22.89 years).
   - For **Punjab**, the test also showed a significant difference ($t = 5.14$, p ¡ 0.001), where urban women's mean age (22.30 years) exceeded that of rural women (21.94 years).

2. **General Linear Regression Model:** Multiple linear regression models were fitted separately for Jammu & Kashmir and Punjab to assess the effects of `v025` (residence), `v106` (education), and `v190` (wealth index) on age at first birth (`age_first_birth`).

   - In **Jammu & Kashmir**, higher education and wealth levels were associated with significantly delayed first births, while rural residence had a negative effect on age at first birth.
   - In **Punjab**, education emerged as the strongest predictor, with "Secondary" and "Higher" education showing large positive coefficients. Wealth effects were smaller but significant for richer groups, while rural women had slightly higher mean ages.

Software and Packages

All statistical analyses were conducted in **R** using the following packages:

- `survey` – for weighted regression and complex sample designs.
- `dplyr`, `ggplot2` – for data manipulation and visualization.
- `broom` – for model output tidying and coefficient extraction.

Interpretation Summary

Both tests and regression results consistently highlight that:

- Urban women tend to have their first birth later than rural women.
- Education is the strongest factor delaying childbearing in both states.
- Wealth contributes positively, but its influence is less pronounced than education.

These findings provide robust evidence of socio-demographic determinants shaping fertility behavior across states.

Determinants and Patterns of Child Survival Under Five Years of Age

Data Cleaning

The rows containing `NA` values in the column `B1$`, the month of birth, are dropped. We assume that the respondents have no children.

Calculation of Dependent Variable

The dependent variable `child-death-under5` is computed from the columns `B5$` and `B7$` in the NHFS-5 dataset. We consider those `CASEID`'s where `B5$` equals zero, and `B7$` is less than 60.

Test of Proportions

A test of equality for proportions was performed to test the null hypothesis that the under five mortality rate is equal for both the states of Punjab and Jammu and Kashmir. The weighted under five mortality by state was computed.

Number of Children Born

A test for equality of proportions was performed to test the null hypothesis that the average number of children born to women in Jammu & Kashmir is equal to the average number of children born to women in Punjab. It was found that the null hypothesis was rejected at a 1% level of significance. The number of children born to women in Punjab is greater and the difference is statistically significant.

Variance Inflation Factor

The Variance Inflation Factor was computed to test for multicollinearity in the second model, testing the relationship of under five child mortality with socioeconomic (wealth index and maternal education level) and environmental and infrastructural factors (area of residence-urban/rural, access to drinking water and sanitation facilities). The results are given in the supplementary tables.

Supplementary Tables

Table 16: Weighted Under-5 Mortality by State (with 95% CI)

State	Weighted Proportion	SE	Lower CI	Upper CI
J&K	0.0150	0.0010	0.0130	0.0169
Punjab	0.0427	0.0016	0.0395	0.0459

Table 17: Comparison of mean number of children ever born in Punjab and Jammu & Kashmir

Statistic	**Value**	**Interpretation**
t-statistic	13.16	Test statistic for mean difference
Degrees of freedom	1591	Based on survey design
p-value	$< 2.2 \times 10^{-16}$	Highly significant difference
95% Confidence Interval	[0.1712, 0.2312]	True mean difference lies within this range
Difference in means	0.2012	Mean number of children (Punjab – J&K)

Note: V201 denotes the total number of children ever born to women aged 15–49. The test is based on a survey-weighted design accounting for complex sampling features of the NFHS-5 dataset.

Table 18: Variance Inflation Factors (VIF) for Multicollinearity Check

Variable	VIF
Wealth Index	1.702275
Maternal Education	1.263585
Area of Residence- Rural/Urban	1.072458
Access to Drinking Water	1.336077
Access to Sanitation Facilities	1.162425
State	1.243535

Influence on employment of women

Data and variables (concise)

Data : NFHS-5 (Individual Recode, IR) subset for Punjab and Jammu & Kashmir. Variables used in the analysis:

- Employment Status: `V714` (Binary) = (1) if working, (0) otherwise.
- Education: `V106` (categorical): No education (0), Primary (1), Secondary (2), Higher (3).
- Wealth: `V190` (categorical) (1) Poorest, ..., (5) Wealthiest.
- Residence: Urban/Rural `V025`.
- Survey design: cluster = V001, strata = V022, sampling weight = V005 (scaled by 10^6 in analysis).

Model specification (compact)

We estimated a design-corrected logistic regression using the `svyglm(..., family = quasibinomial)` framework:

$$\text{logit}\big[P(\text{Employment Status} = 1)\big] = \beta_0 + \beta_1\text{Education} + \beta_2\text{Wealth} + \beta_3\text{Residence} + \beta_4\text{State}.$$

All the inference accounts for clustering, stratification and sampling weights (Taylor linearization variance).

Work allocation

Roll No.	Name	Contribution
BS2306	Aishee Bhattacharya	Effect of education on fertility rate of women
BS2323	Bipayan Banerjee	Early Marriage in India: Prevalence and Predictors
BS2333	Nandita Sen	Urban–Rural Divide in Age at First Birth: The Role of Education and Wealth
BS2338	Rahul Konar	Determinants and Patterns of child survival
BS2357	Srijeet Bhattacharjee	Influence on employment of women